Question	Answer	Marks
5(a)	sound waves are longitudinal / not transverse (and only transverse waves can be polarised)	**B1**
5(b)(i)	A periodic curve of at least one period with minimum 0 and maximum I_0	**B1**
	Smooth continuous $\sin^2\alpha$ curve	**B1**
	Peaks at 90° and 270° **and** troughs at 0, 180° and 360°	**B1**
5(b)(ii)	$I \propto A^2$	**C1**
	$I = I_0 \cos^2 \theta$	**C1**
	At $\alpha = 20°$ the angle between the planes of polarisation of light and filter is 70° (or 110°) $I = I_0 \cos^2 70$ **or** $I = I_0 \cos^2 110$ ($I = 0.12\ I_0$)	**C1**
	amplitude = $0.34\ A_0$	**A1**

Question	Answer	Marks